

PHYS 4027 / 5027 • FALL 2026 • LECTURE 12

Boundary Value Problems



Prof. Alex Chen

Department of Physics

Washington University in St. Louis

Two ways to pin down a second-order ODE



A second-order ODE, $y'' = f(x, y, y')$, needs two conditions.

- **Initial value problem.** Both at the same point: $y(a)$ and $y'(a)$. Everything so far.
- **Boundary value problem.** One at each end: $y(a) = \alpha$ and $y(b) = \beta$.
 - A ball thrown from the hand that has to go through the hoop.
 - The electric potential between two electrodes held at fixed voltages.
 - A quantum particle in a box: $\psi = 0$ at both walls.

An IVP can be marched forward one step at a time. A BVP cannot: a naive marching cannot guarantee the solution will meet $y(b) = \beta$.

Part I · Shooting



Find the missing initial condition

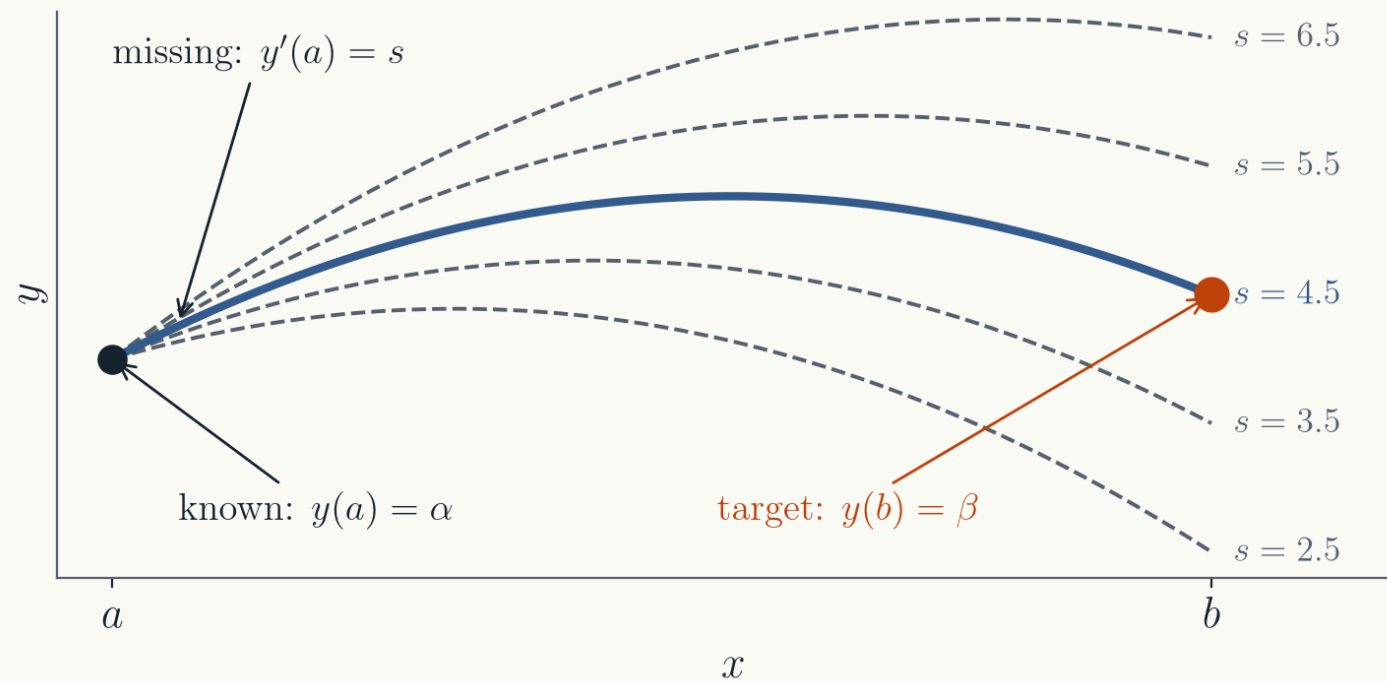
Shooting a basket ball

How do you find out the way to shoot a projectile, so that it lands at a target location?



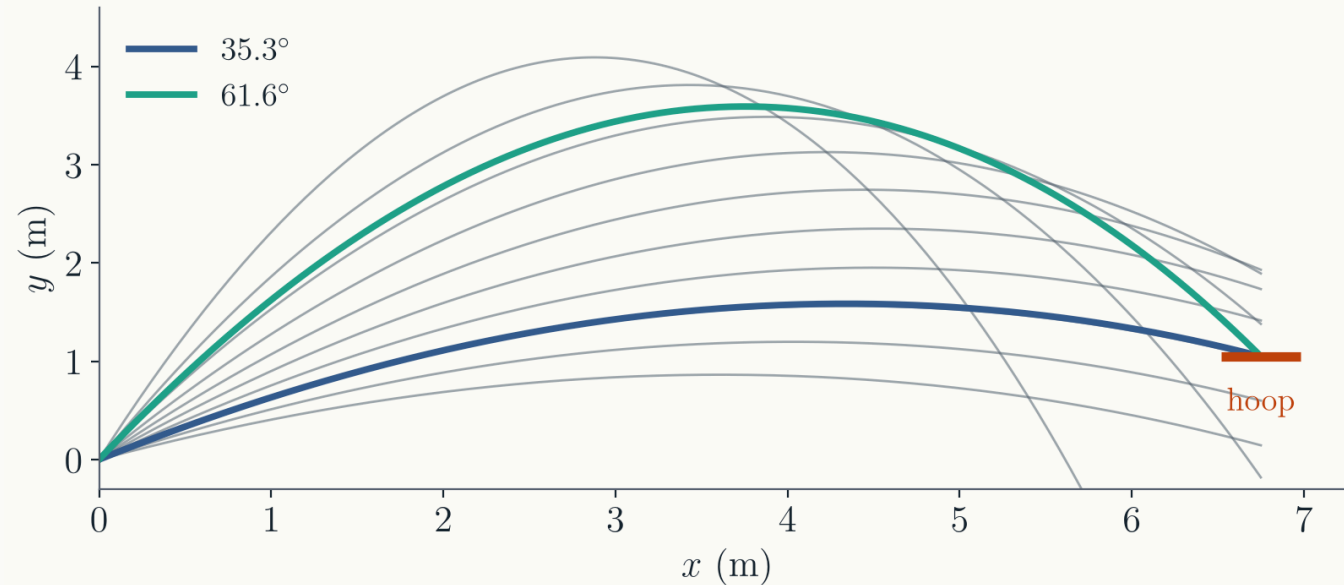
The missing initial condition

$y'' = f(x, y, y')$ with $y(a) = \alpha$ and $y(b) = \beta$. To march it like an IVP, we need $y(a)$ and $y'(a)$. We have only the first.



Shooting: find the initial slope $y'(a)$ whose solution lands on $y(b) = \beta$.

Shooting a basketball



The ball leaves the hand at 10 m/s, with air drag. The hoop is 6.75 m away and 1.05 m higher.

The starting point and the speed are known. The missing initial condition is the angle.

Try an angle, follow the ball to the hoop, see by how much it misses, and adjust.

Two angles go in: a flat shot and a high arc.

What we are solving



Call the missing slope s , and solve the IVP

$$y'' = f(x, y, y'), \quad y(a) = \alpha, \quad y'(a) = s.$$

The solution $y(x; s)$ depends on s , and so does where it ends up at $x = b$. Measure the miss:

$$F(s) = y(b; s) - \beta$$

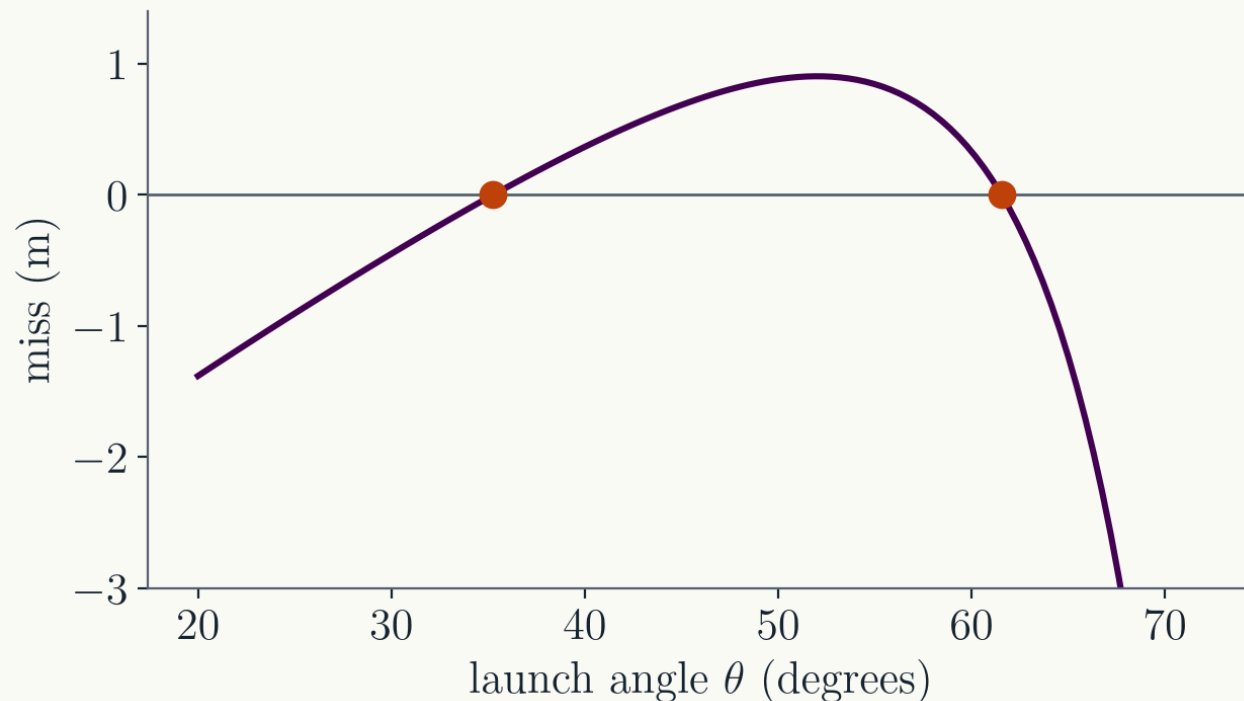
Shooting is solving $F(s) = 0$: one equation for one unknown.

One evaluation of $F(s)$:

1. set $y(a) = \alpha$ and $y'(a) = s$;
2. integrate to $x = b$ with an IVP solver (Homework 5's Dormand-Prince, landing on b exactly);
3. return $y(b) - \beta$.

For the basketball, s is the launch angle and F is how far above or below the hoop the ball passes.

Solving $F(s) = 0$



F has no formula, but we can evaluate it at any s . Any root finder from Lecture 3 works. The secant method needs no derivative:

$$s_{n+1} = s_n - F(s_n) \frac{s_n - s_{n-1}}{F(s_n) - F(s_{n-1})}$$

Each iteration is one new shot. Starting from 30° and 35° : 35.3° after 6 shots. Starting from 60° and 65° : 61.6° after 9.

F has two roots, so the problem has two solutions. The starting guesses decide which one you get.

Eigenvalue problems



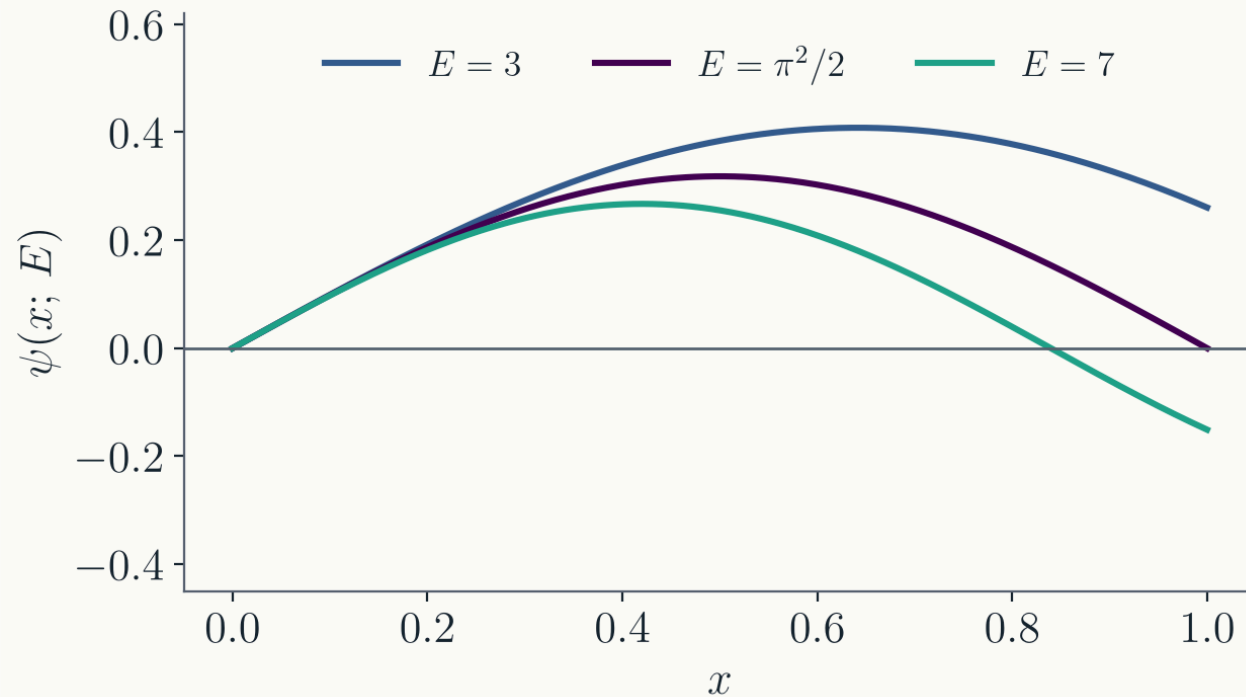
The time-independent Schrödinger equation for a particle of mass m in a potential $V(x)$:

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + V(x) \psi = E \psi$$

- The infinite well: $V = 0$ for $0 < x < L$, and $V = \infty$ outside. The particle cannot leave the box, so ψ must vanish at the walls.
- In units with $\hbar = m = L = 1$:

$$-\frac{1}{2} \psi'' = E \psi, \quad \psi(0) = 0, \quad \psi(1) = 0$$

What to shoot with



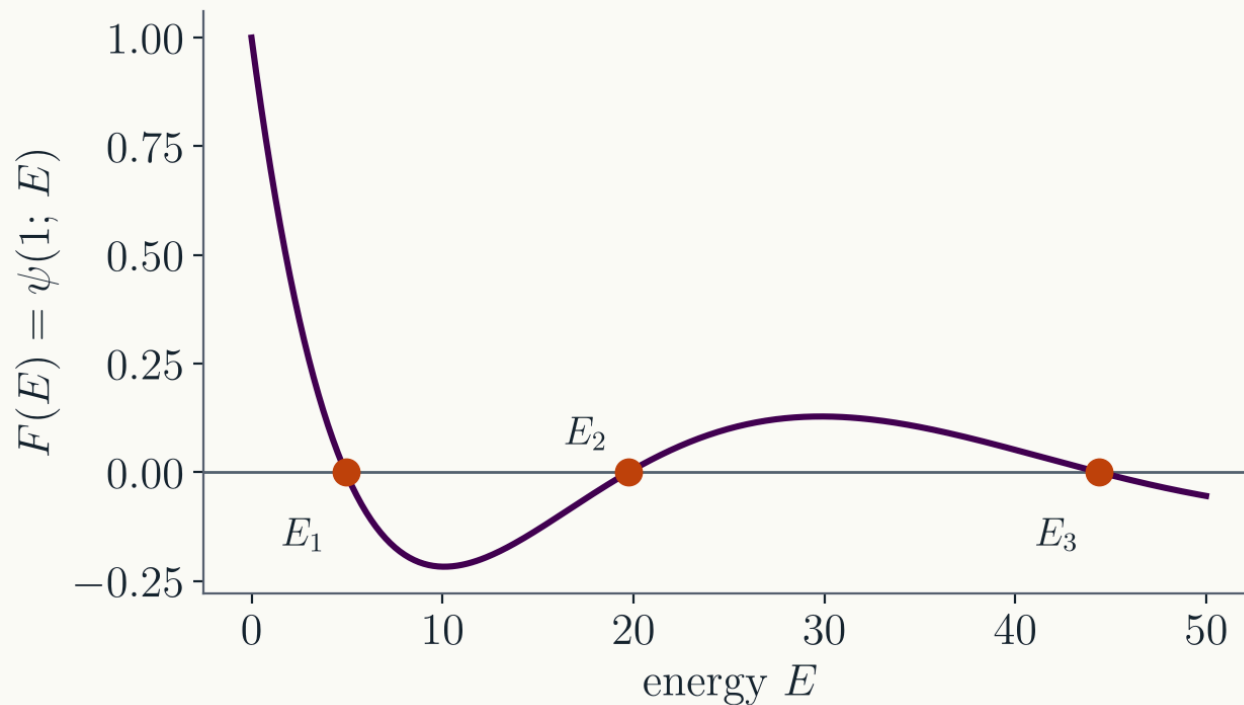
At $x = 0$ we know $\psi(0) = 0$. Two things are unknown: the slope $\psi'(0)$ and the energy E .

The slope only sets the size: doubling $\psi'(0)$ doubles ψ everywhere, and the zeros stay where they are. So fix $\psi'(0) = 1$.

The energy changes the shape. Too low, and ψ has not come back to zero by $x = 1$; too high, and it crosses zero too early.

So we fix $\psi'(0) = 1$ and shoot in E .

Root finding in E



Same recipe, with E in place of s :

$$F(E) = \psi(1; E) = 0$$

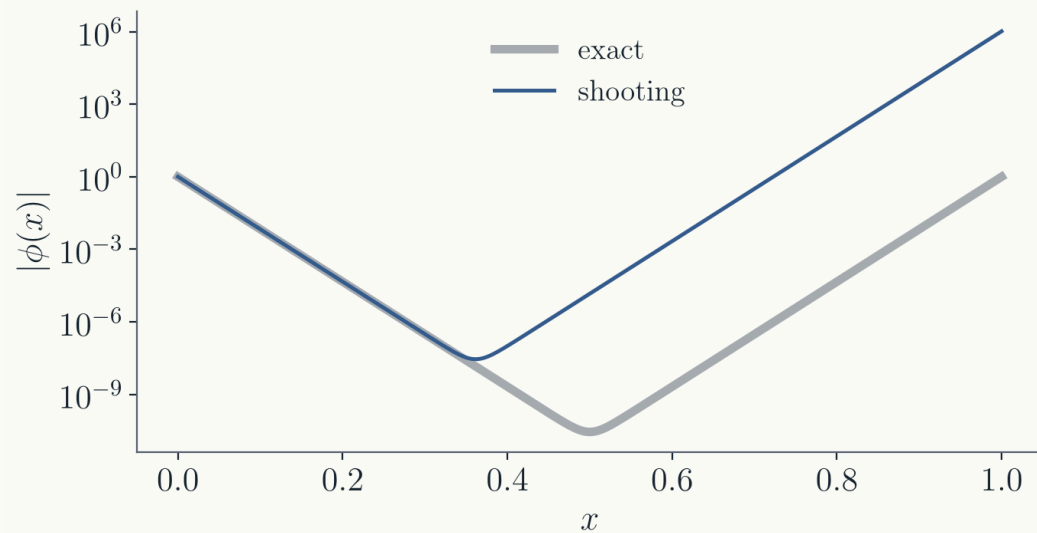
Each root is an energy level. The secant method started near the first one finds $E_1 = 4.9348022005$, equal to $\pi^2/2$ to 10 digits, in 9 IVP solves.

Last step: rescale ψ so that $\int_0^1 \psi^2 dx = 1$.

When shooting fails

A linear BVP with a known exact solution:

$$\phi'' = \kappa^2 \phi, \quad \phi(0) = \phi(1) = 1 \quad \Longrightarrow \quad \phi(x) = \frac{\cosh \kappa(x - \frac{1}{2})}{\cosh(\kappa/2)}$$



Every solution is a mix of $e^{-\kappa x}$ and $e^{+\kappa x}$. The exact one decays away from each end over a length $1/\kappa$, down to $e^{-\kappa/2} \approx 10^{-11}$ in the middle for $\kappa = 50$.

Shooting from $x = 0$, an error δs in the slope adds $\frac{\delta s}{\kappa} \sinh(\kappa x)$: a little of $e^{+\kappa x}$, which grows by $e^{\kappa} \approx 5 \times 10^{21}$ on the way across.

Here the slope is right to 15 digits, and the solution still misses by a factor of 10^6 .

Shooting marches across the whole domain. We need a method that does not march.

Part II · The grid method



Solve for every point at once

The idea



- Put a grid on $[a, b]$: $x_i = a + ih, i = 0, \dots, N$, with $h = (b - a)/N$. The unknowns are all the values y_i at once.
- Replace every derivative in the ODE by a difference of neighbouring y_i . Each interior point gives one equation.
- The boundary conditions are simply $y_0 = \alpha$ and $y_N = \beta$. Both ends are imposed exactly, and nothing is integrated from one end to the other.
- The ODE becomes $N - 1$ algebraic equations for $N - 1$ unknowns. Numerical Recipes calls this the *relaxation* method; it is also called the *finite-difference* method.

Finite differences

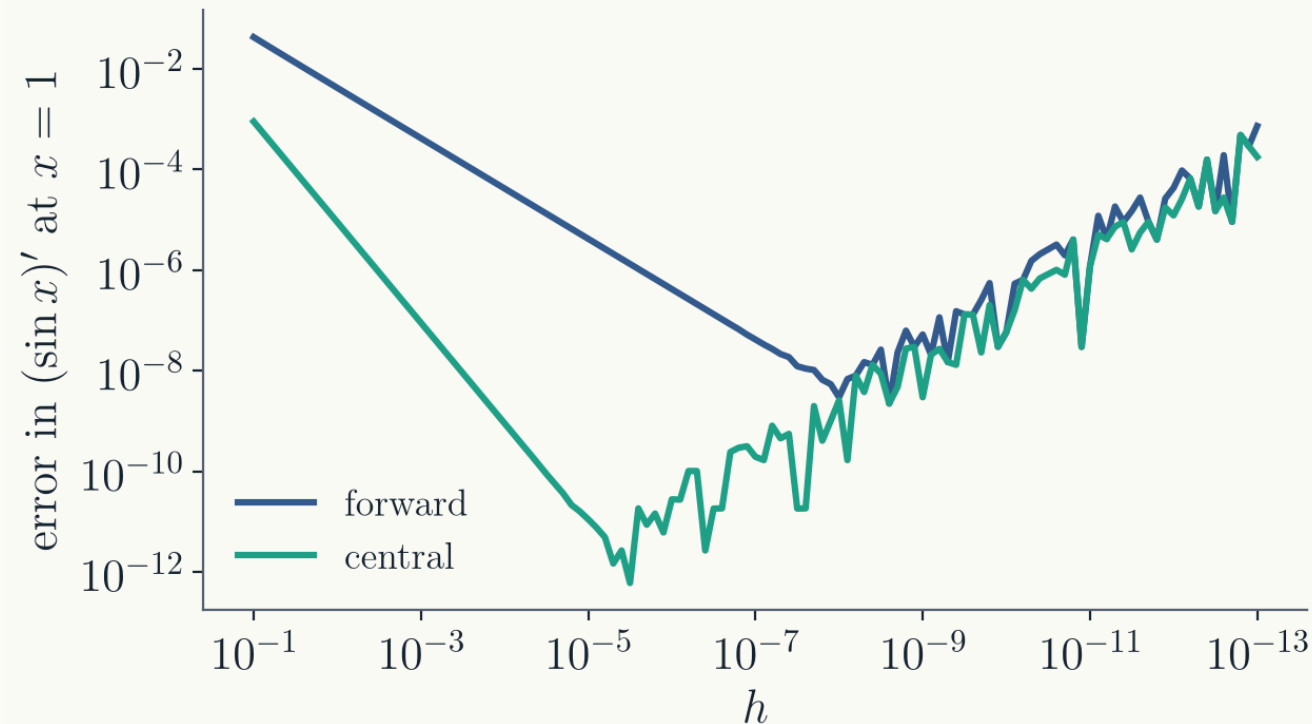


Taylor-expand the neighbours, $y_{i\pm1} = y_i \pm hy'_i + \frac{h^2}{2}y''_i \pm \frac{h^3}{6}y'''_i + \dots$, and combine:

forward	$y'_i \approx \frac{y_{i+1} - y_i}{h}$	error $-\frac{h}{2}y''$	first order
central	$y'_i \approx \frac{y_{i+1} - y_{i-1}}{2h}$	error $-\frac{h^2}{6}y'''$	second order
second derivative	$y''_i \approx \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2}$	error $-\frac{h^2}{12}y''''$	second order

Symmetric formulas cancel the even (or odd) terms of the expansion and gain an order. More neighbours give higher order.

The order, and the rounding floor



The derivative of $\sin x$ at $x = 1$. Truncation error falls with slope 1 and 2.

Below $h \approx 10^{-8}$ (forward) and 10^{-5} (central), the subtraction $y_{i+1} - y_i$ cancels most digits, and the rounding error grows like ϵ/h : catastrophic cancellation, Lecture 1.

A BVP grid has $h \sim 10^{-3}$ or larger, far from the floor. There, the order is what matters.

The same problem on a grid

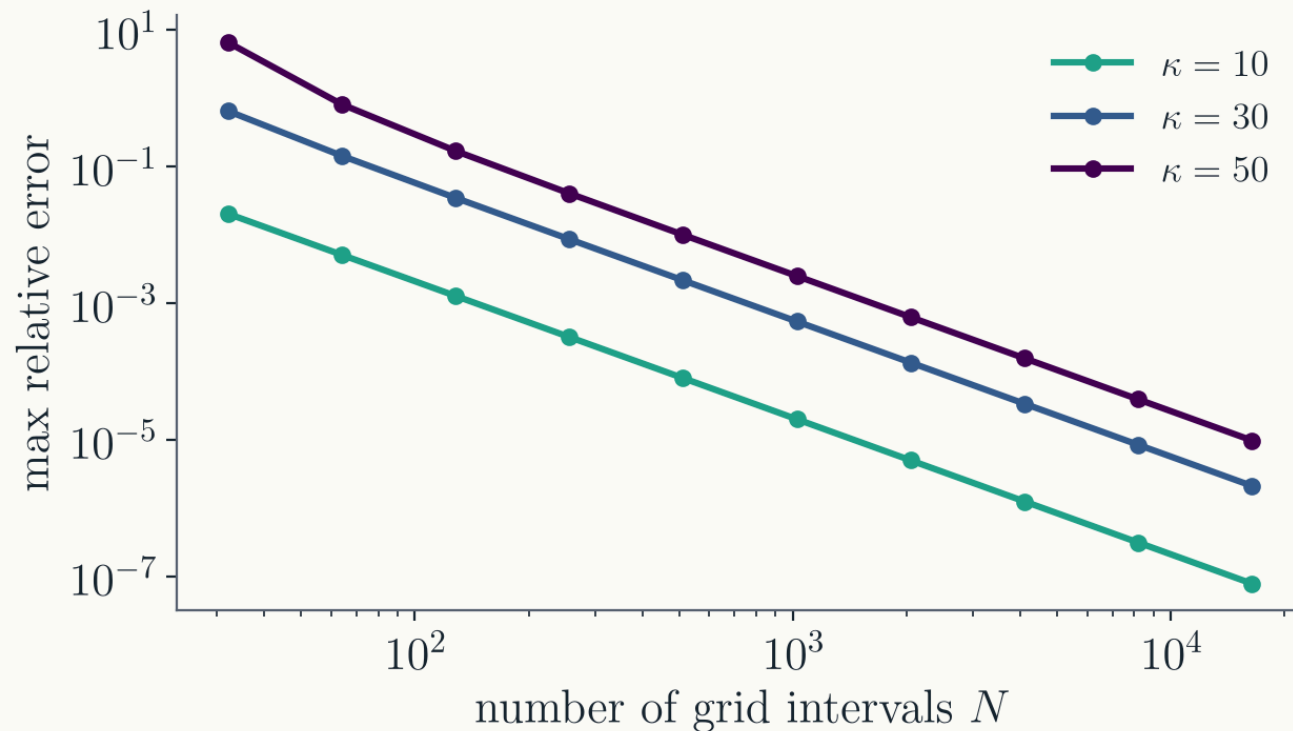
At each interior point, $\frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h^2} - \kappa^2 \phi_i = 0$. The known ϕ_0 and ϕ_N move to the right-hand side:

$$\frac{1}{h^2} \begin{pmatrix} -2 - \kappa^2 h^2 & 1 & & & \\ & 1 & -2 - \kappa^2 h^2 & \ddots & \\ & & \ddots & \ddots & 1 \\ & & & 1 & -2 - \kappa^2 h^2 \end{pmatrix} \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_{N-1} \end{pmatrix} = -\frac{1}{h^2} \begin{pmatrix} \phi_0 \\ 0 \\ \vdots \\ \phi_N \end{pmatrix}$$

A linear ODE becomes one linear system, $\mathbf{A}\phi = \mathbf{b}$. Each row involves only a point and its two neighbours, so $\mathbf{A}$ is *tridiagonal*.

Solve it once, and every ϕ_i comes out together. No guessing, no iteration.

The grid succeeds where shooting failed



Slope 2 for every κ , as the second-order formula promises.

At $\kappa = 50$ with $N = 1024$: relative error 2.5×10^{-3} , where shooting was off by a factor of 10^6 .

Larger κ needs a finer grid, since h has to resolve the decay length $1/\kappa$. But the error is controlled by h , and no longer amplified by e^κ .

A nonlinear version



Make the same problem nonlinear, $\phi'' = \kappa^2 \sinh \phi$, and use the same grid. One equation per interior point:

$$\frac{\phi_{i+1} - 2\phi_i + \phi_{i-1}}{h^2} - \kappa^2 \sinh \phi_i = 0, \quad i = 1, \dots, N - 1$$

- Still $N - 1$ equations for $N - 1$ unknowns, and each one involves only a point and its two neighbours.
- But $\sinh \phi_i$ is not linear in ϕ_i , so this is no longer a matrix equation $\mathbf{A}\phi = \mathbf{b}$ that we can solve in one go.
- Call the left-hand side $F_i(\phi)$. We need $\mathbf{F}(\phi) = 0$: root finding again, now for a vector of unknowns.

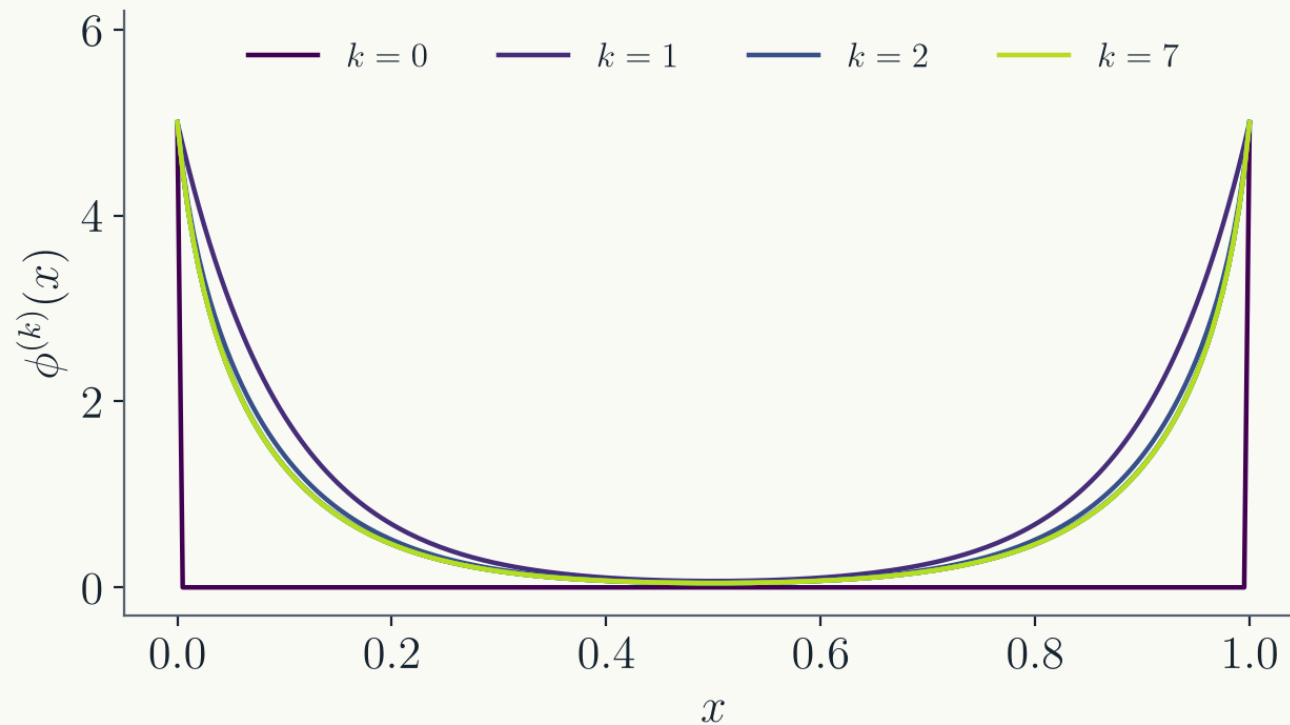
Newton's method for many unknowns

	one unknown (Lecture 3)	a vector of unknowns
linearize	$F(x + \delta) \approx F(x) + F'(x) \delta$	$\mathbf{F}(\boldsymbol{\phi} + \boldsymbol{\delta}) \approx \mathbf{F}(\boldsymbol{\phi}) + \mathbf{J} \boldsymbol{\delta}$
step	$\delta = -F/F'$	solve $\mathbf{J} \boldsymbol{\delta} = -\mathbf{F}$
update	$x \leftarrow x + \delta$	$\boldsymbol{\phi} \leftarrow \boldsymbol{\phi} + \boldsymbol{\delta}$

- $\mathbf{J}$ is the Jacobian, the matrix of derivatives $J_{ij} = \partial F_i / \partial \phi_j$. It plays the role of F' .
- F_i depends only on $\phi_{i-1}, \phi_i, \phi_{i+1}$, so $\mathbf{J}$ is tridiagonal, just like the matrix of the linear problem: $1/h^2$ next to the diagonal, $-2/h^2 - \kappa^2 \cosh \phi_i$ on it.

Each Newton iteration is one tridiagonal linear system. A nonlinear BVP becomes a sequence of linear ones.

Newton converges quickly



$\kappa = 10$, $\phi = 5$ at both ends, $N = 200$, starting from $\phi = 0$ inside.

k	$\max \delta $
1	4.8
2	0.61
3	0.15
4	6.5×10^{-3}
5	1.0×10^{-5}
6	2.4×10^{-11}

Once close, the number of correct digits doubles each iteration.

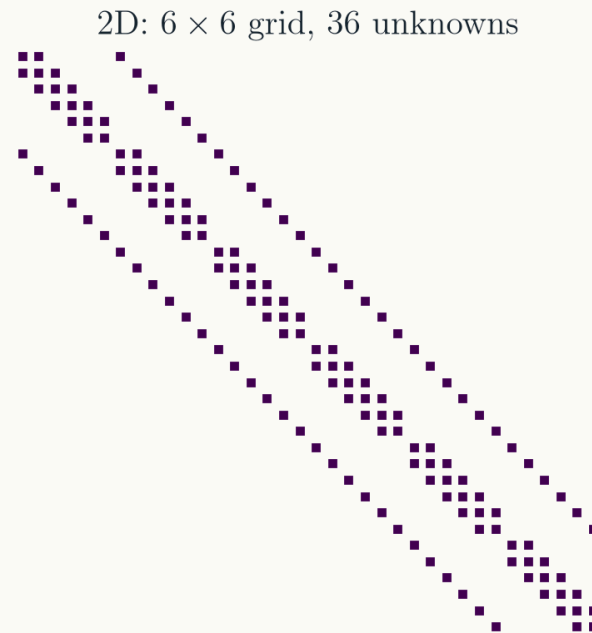
Shooting or grid?



	shooting	grid
reuses	your IVP solver and a root finder	finite differences and a linear solver
accuracy	adaptive, set by the tolerance	set by h : second order here
weak spot	growing solutions amplify errors across the domain	needs a good initial guess when nonlinear
many unknown functions	a multidimensional root find	a larger, still sparse, system

Shooting is quick to write when the problem is benign. The grid method is the one that carries over to partial differential equations.

Every grid method ends in $\mathbf{Ax} = \mathbf{b}$



In 2D, each point couples to four neighbours. A 6×6 grid gives a 36×36 matrix with 156 nonzeros out of 1296.

A 1000×1000 grid has 10^6 unknowns. How do we solve that system, and how fast?

The square well on a grid becomes $\mathbf{A}\psi = E\psi$: a matrix eigenvalue problem.

Next two weeks: numerical linear algebra.